

MSc THESIS PROPOSAL

Interuniversity Program in Water Resources Engineering

Evaluating Satellite-Based Rainfall Estimates for Flood Modelling and Early Warning *in the Ky Lo River Basin, Vietnam*

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Floods: A Global & Regional Crisis

90%

of all natural disasters
are water-related

54%

of WRDs are floods
most frequent disaster

41%

of global flood events
occur in Asia

75%

of river basins globally
have inadequate gauge
networks

Study Area: Ky Lo River Basin, Vietnam

Basin Area

2,058 km²

**Basin Area &
Main River Length**

2,058 km²
103 km

Rainy Season

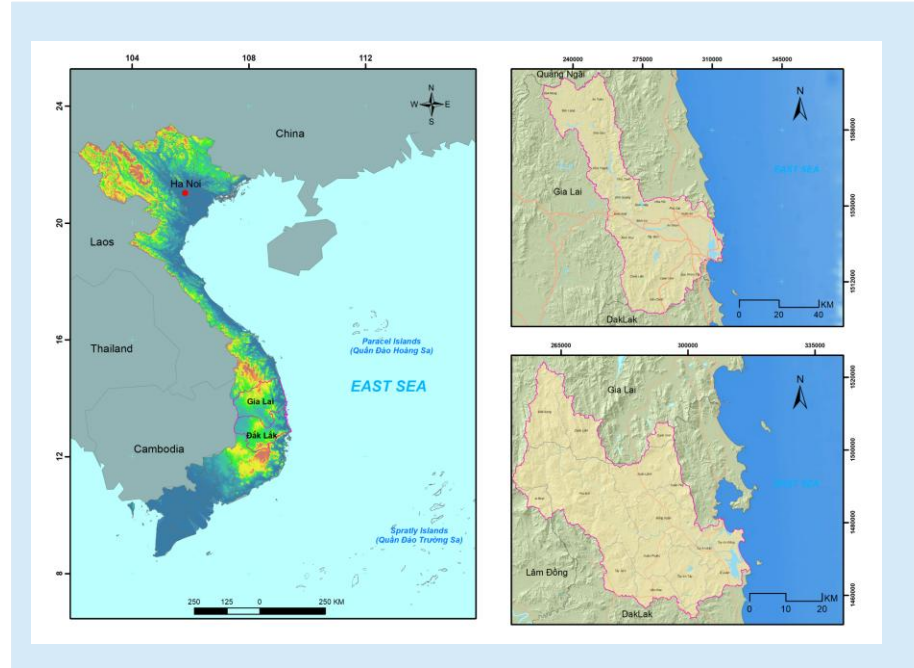
Oct – Nov (85% of floods)

Gauge Availability

Sparse gauges cannot capture
high spatial-temporal rainfall
variability

Climate Change

+9,562 ha flooded by 2070



Research Gap & Motivation

Literature

Performance of Satellite Precipitation Estimates (SPEs)

- GPM IMERG Final Run (IMERGF-V6) shows the best overall performance for Vietnamese basins in rainfall detection and streamflow simulation (Le et al., 2020).
- CHIRPS exhibits the lowest bias, making it suitable for long-term water resources planning (Le et al., 2020).
- CHIRPS and PERSIANN-CDR demonstrate reliability for long-term trend analysis in Northern Vietnam (Nguyen et al., 2025).

SPEs vs. Gauge Data in Hydrological Modeling

- In low-gauge-density basins, SPE-driven SWAT models outperform gauge-driven simulations, highlighting SPE value in data-scarce regions (Le et al., 2020).
- Bias correction substantially improves SPE-driven hydrological simulations, especially when integrated with physically based models like SWAT (Belay et al., 2019).

Knowledge Gap

- No study has evaluated near-real-time SPEs for operational Flood Early Warning System in the Ky Lo basin
- Lead-time feasibility (satellite latency vs. flood travel time) is unquantified
- SWAT vs. HEC-HMS for extreme floods in this basin has not been compared
- Rainfall thresholds for local early warning activation do not exist
- A replicable SPE to Flood Early Warning System framework for data-scarce Vietnamese basins is lacking

Research Questions

1

Which near-real-time SPE product (IMERG Early Run, CHIRP, or PERSIANN) best represents rainfall in the Ky Lo Basin compared to ground observations?

2

To what extent can bias correction improve satellite rainfall performance for hydrological simulation?

3

How accurately do SWAT and HEC-HMS capture extreme flood peaks when driven by bias-corrected SPEs, and are they suitable for early warning applications?

4

Is there sufficient lead time from near-real-time SPEs to support an operational flood early warning system?

Study Objectives



Objective 1

Extract & Compare SPEs:

Harmonize IMERG Early Run, CHIRP, and PERSIANN using Google Earth Engine and GIS



Objective 2

Validate & Bias-Correct:

Validate against gauges (2021–present) and apply Linear Scaling & Quantile Mapping



Objective 3

Comparative Modelling:

Set up and calibrate SWAT and HEC-HMS, evaluate accuracy of both models for extreme flood simulation and assess early warning suitability



Objective 4

Early Warning Systems Feasibility:

Evaluate satellite latency vs. flood travel time

Methodology Framework

01



Data Acquisition & Harmonization

GEE + Python automation
IMERG / CHIRP / PERSIANN
DEM, Land Use and Land
Cover, Soil data

02



Statistical Performance Assessment

RMSE, CC, Relative Bias
Linear Scaling
Quantile Mapping

03



Hydrological Modelling

SWAT (SUFI-2 calibration)
HEC-HMS (event-based)
NSE, KGE, PBIAS

04



EWS Feasibility Evaluation

Lead-time analysis
4 hr latency vs. 6–10 hr
Rainfall thresholds

Satellite Precipitation Products (SPEs)



PRIMARY [Near Real-Time]

GPM IMERG Early Run

Specifications

- Resolution: 0.1° spatial
- Temporal: 30-minute
- Latency: ~4 hours
- Version: V06

Main product for EWS, only SPE with sufficient lead time for operational warning



BENCHMARK [Low Bias]

CHIRP V2.0

Specifications

- Resolution: 0.05° (finest)
- Temporal: Daily
- Latency: Several days
- Version: V2.0

Used as bias reference and long-term benchmark, not suitable for real-time EWS



CANDIDATE [Hourly]

PERSIANN

Specifications

- Resolution: 0.25°
- Temporal: Hourly
- Latency: ~1 day
- Version: Standard

Near-real-time candidate, mixed performance in Vietnam which requires local validation

Statistical Performance Assessment & Bias Correction

Rainfall Accuracy Metrics

Evaluated against ground-based gauge observations

RMSE Root Mean Square Error

Error magnitude between satellite and gauge data. Sensitive to large errors, critical for detecting failures in high-intensity events.

CC Pearson Correlation Coefficient

Temporal agreement between satellite and gauge time series, how well each product reproduces rainfall timing and variability.

RB Relative Bias

Systematic directional error (over/underestimation) that propagates into model inputs and degrades flood simulation accuracy.

Bias Correction

Systematic biases in SPE products will be corrected using available ground-based gauge data prior to hydrological model integration. Bias correction methods will follow approaches demonstrated to significantly improve hydrological simulations in data-scarce regions

The best-performing corrected dataset will be used as model input.

Hydrological Performance Metrics

Applied to simulated streamflow vs. observed discharge after bias correction

NSE Nash-Sutcliffe Efficiency

Predictive skill vs. the observed mean. Sensitive to peak flow errors — the most critical dimension for flood simulation.

KGE Kling Gupta Efficiency

Decomposes model error into correlation, variability ratio, and bias ratio. more balanced assessment than NSE alone.

PBIAS Percent Bias

Volumetric over/underprediction across the full simulation period; identifies systematic mass balance errors.

RSR RMSE Obs. Std. Dev. Ratio

Normalises RMSE by observed variability, enabling comparison across sub-catchments with different flow regimes.

Hydrological Model Implementation

SWAT

Soil and Water Assessment
Tool

Continuous

Calibration Method

SUFI-2 (Sequential Uncertainty Fitting) algorithm within SWAT-CUP, following established protocols.

Core Strength

Widely validated across Vietnamese basins. Satisfactory simulations using satellite rainfall in the Srepok River Catchment, a comparable data-scarce setting.

HEC-HMS

Hydrologic Engineering Center -
HMS

Event-based

Calibration Method

Calibrated against observed discharge and tested against historical flood event data to evaluate real flood condition accuracy.

Core Strength

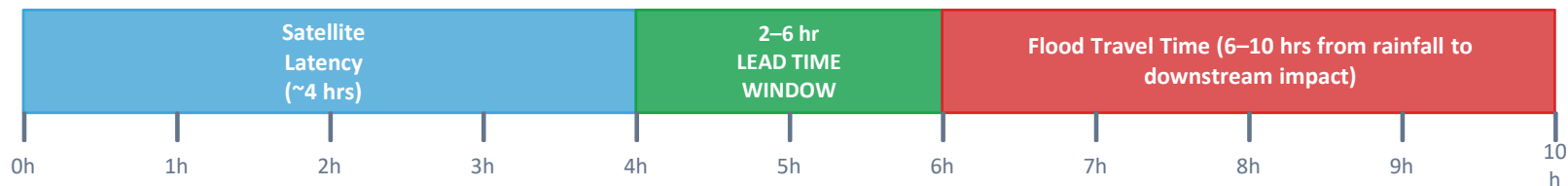
Stronger for Extreme Event Simulation

→ Forecast Period: 2021-2023 simulated vs. observed gauge data

→ Both models evaluated for accuracy with each model tested in EWS simulation to assess early warning utility and threshold derivation

Early Warning System: Lead-Time Analysis

Key Insight: GPM IMERG Early Run (~4 hr latency) offers a 2-to-6-hour warning window before downstream flood impact (6–10 hr travel time). This study will quantify whether this window is sufficient for effective warning dissemination and community response in the Ky Lo Basin.



IMERG Early Run available

~4 hr latency

Warning dissemination window

2 to 6 hour lead time

Flood arrival at downstream

6–10 hrs travel time

Expected Outcomes & Contributions



Validated SPE Datasets

Bias-corrected IMERG, CHIRP & PERSIANN datasets ready for operational use in the Ky Lo Basin



Calibrated Hydro Models

SWAT and HEC-HMS models with comparative performance benchmarks for extreme flood simulation



Rainfall Thresholds

Site-specific Early Warning System activation thresholds for downstream flood warning in Phu Yen Province



EWS Lead-Time Assessment

Quantitative evidence on whether near-real-time SPEs can support actionable warnings



Replicable Framework

Methodology transferable to other data-scarce basins across Vietnam and Southeast Asia



Scientific Contribution

Satellite Precipitation Estimates to Flood Early Warning Sign evaluation for the Ky Lo Basin

Research Timeline (July 2026 - May 2027)

Task	26-Jul	26-Aug	26-Sep	26-Oct	26-Nov	26-Dec	27-Jan	27-Feb	27-Mar	27-Apr	27-May
Literature Review											
Data Acquisition and Pre-processing											
1. Acquire satellite rainfall datasets and gauge data											
2. Data cleaning, harmonization, and GIS processing											
3. DEM, land use, and soil data preparation											
Satellite Rainfall Validation and Bias Correction											
1. Statistical validation against rain gauges											
2. Apply and evaluate bias correction methods											
Hydrological Model Development											
1. SWAT model setup and calibration											
2. HEC-HMS model setup and calibration											
3. Model validation and comparative assessment											
Flood Forecasting and Early Warning Evaluation											
1. Flood simulation using SPE products											
2. Lead-time and EWS performance evaluation											
3. Development of recommended rainfall thresholds											
Writing of Thesis and Final Revisions											
1. Drafting of methodology and results chapters											
2. Final analysis, discussion, and conclusions											
3. Final editing, revision, and defense preparation											

Thank You

Questions & Discussion



3 SPE
Products



2 Hydro
Models



Early Warning
Systems



Replicable
Framework